AI-Guided Application Deployment: CodeArts via KooCLI



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AI-Assisted Migration: Usage Guidelines

Practical guidance, not a formal delivery commitment

How to work with AI for VM-to-ECS migration

How to Use This Guide

This guide shows what to ask your AI assistant, what knowledge it needs, and how to verify its work to prevent errors. AI accelerates migration planning and execution -- but does not replace expert review or customer decisions.

1. What AI Does

Planning / Analysis / Automation

- AI plans, analyzes, generates scripts, and assesses risks
- It does not replace design, expert review, testing, or decisions
- Review all outputs before acting on them

2. What to Provide

Anonymized Data Only

- Use simulated, test, or anonymized data only
- Never enter real credentials, passwords, keys, or customer data
- Respect security, compliance, and confidentiality

3. How to Work with AI

Review / Authorize / Verify

- Review every AI output before acting
- Authorize each migration step explicitly
- Keep a human in the loop for all production changes

4. When to Be Cautious

Sensitive Systems / Manual Review

- For sensitive systems (financial, defense, classified), consider manual migration
- Use strict AI oversight for compliance-heavy workloads
- Always prioritize security and sovereignty requirements

AI-assisted migration improves efficiency; it does not replace human judgment. Verify every step against real data.

Your CodeArts Deployment Scenario

Your Starting Point

You have a growing application portfolio built and released by hand across several environments. Builds are inconsistent and deployments are slow. You need a repeatable, automated delivery pipeline.

Your Goal

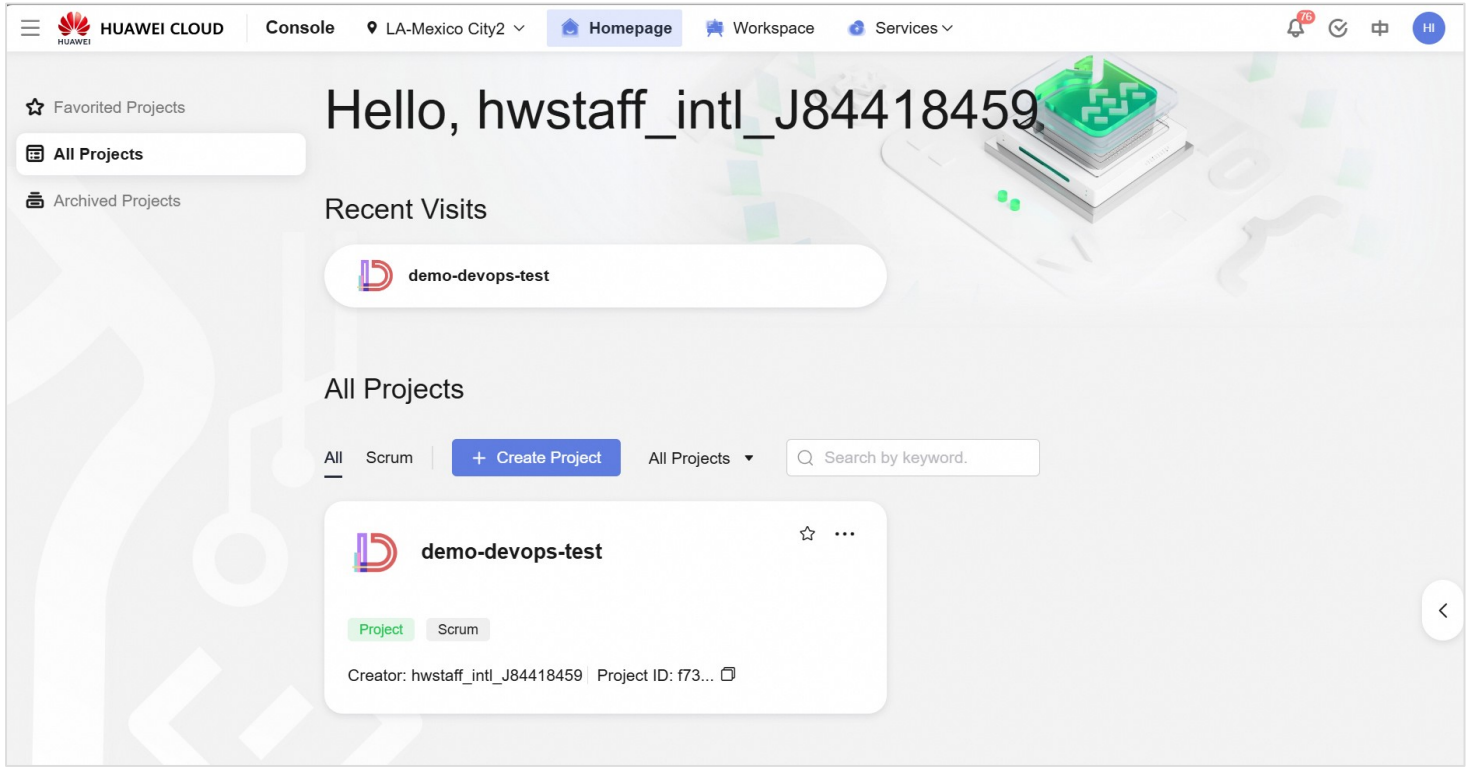
Stand up automated CI/CD with CodeArts Repo, Build, and Deploy, orchestrated through Pipeline and KooCLI. Ship faster with fewer errors. Keep every environment consistent.

Why AI-Assisted

AI can scan your repositories in minutes, map pipeline dependencies automatically, and generate the KooCLI commands to orchestrate the deployment. What used to take days of manual setup now takes hours.

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What CodeArts Looks Like



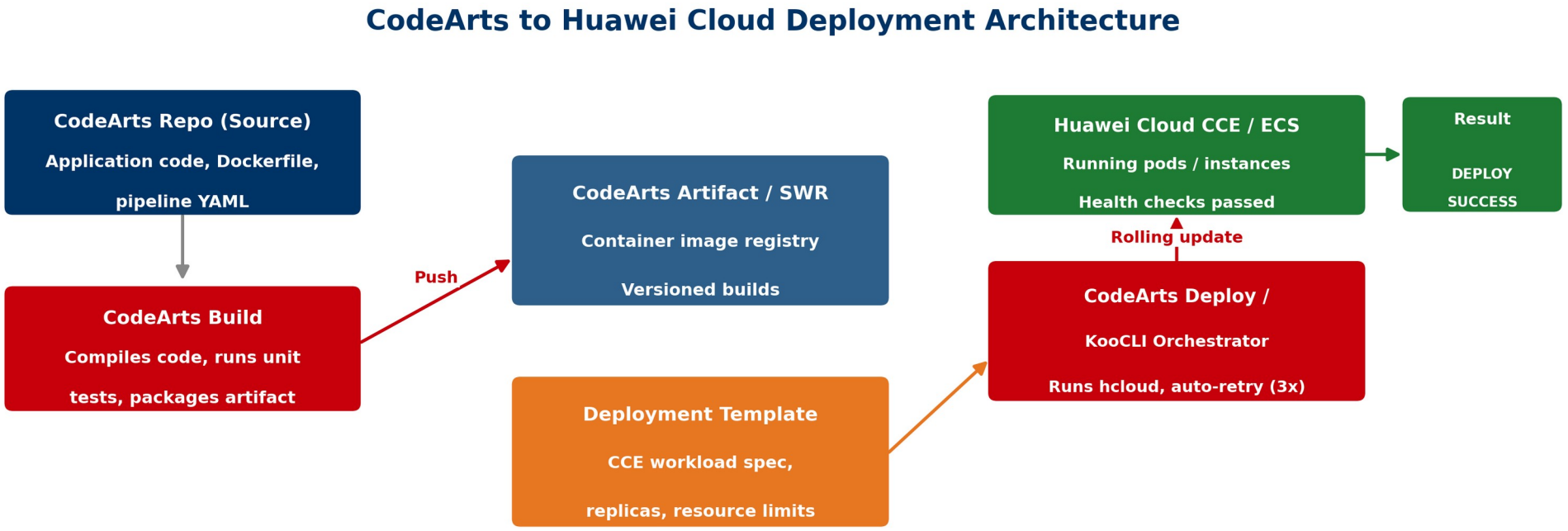
Huawei Cloud CodeArts Console — Homepage view (Project: demo-devops-test)

Console at a Glance

- Console ▸ Workspace ▸ Homepage navigation is shared across every Huawei Cloud service, including CodeArts
- Each project exposes a Project ID (e.g. f73...) — the identifier CodeArts Pipeline, Build, and Deploy APIs expect, and the one KooCLI commands reference
- This project follows the Scrum template; Kanban and custom workflows are also available per project
- The region selector (e.g. LA-Mexico City2) determines where pipelines, build agents, and deployment targets run
- Recent Visits and Favorited/Archived Projects keep active repositories and pipelines one click away
- The Services menu links directly to Repo, Pipeline, Build, Deploy, and the rest of the CodeArts suite

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Step 1: Ask AI to Discover Your CodeArts Project



Example prompt: "Review my CodeArts project and list all repositories, pipelines, and how each service depends on the others"

AI Should Know: How to read your CodeArts Repo structure, identify which pipeline stages depend on which build artifacts, and group them by application

Verify: Compare the AI's list against your actual CodeArts project. Make sure no repositories or pipelines are missing and all dependencies are captured

Prevent Hallucinations: The AI should reference real repository and pipeline names from your project, not make them up

Source: Huawei Cloud CodeArts Documentation (https://support.huaweicloud.com/codearts/). — Pipeline Service Overview. Accessed: Aug 2026.

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Step 2: Ask AI to Prepare Deployment Pipelines

What to Ask AI

Example prompt: "Prepare a CodeArts Pipeline for each service. Choose target CCE workload sizes that match or exceed what we run today. Set up networking and security for the new environment."



AI Should Know & Verify

The available CCE node flavors and ECS instance types on Huawei Cloud, how to map build artifacts to container images, and how to configure networks securely.
Verify: Each planned workload is properly sized -- not too small, not wastefully large. AI should use real options from the catalog, not guesses.



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Step 3: Ask AI to Execute Deployment

Deployment Job Execution: Two-Phase Process

Phase 1: Build & Package

- CodeArts Build compiles code and runs tests
- Non-disruptive — current version stays live
- Build speed depends on project size and cache
- Typical build: 3–8 min per service
- Monitor: build %, test results, artifact size

Ready →

Phase 2: Rolling Deploy

- Brief rollout window: 1–3 min per service
- Health checks before traffic switch
- KooCLI triggers CCE/ECS deployment APIs
- Switch traffic to new revision
- Roll back automatically on failed health check



On failure: review logs → fix issue → retry from last checkpoint

Source: Huawei Cloud CodeArts Pipeline Documentation (https://support.huaweicloud.com/codearts/) — CI/CD Workflow. Accessed: Aug 2026.

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Step 4: Ask AI to Scale and Handle Errors

What to Ask AI

Example prompt: "Scale up and deploy the remaining services in batches, grouped by dependency. Start with the least critical ones first. Handle any errors automatically."



AI Should Know

Common deployment issues and how to fix them. When a rollout won't succeed one way, try an alternative approach. How to pause and resume if needed.



Verify & Prevent Hallucinations

When the AI says an error is 'fixed', verify it yourself. The AI should explain what went wrong and what it did, not just say 'resolved'. Never assume a retry succeeded without checking.



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Step 5: Ask AI to Validate Every Deployment

What to Ask AI

Example prompt: "Check every deployed service -- does it start? Is the network working? Are all health checks passing? Give me a report with specific results for each service."



AI Should Know

What a successful deployment looks like, how to compare metrics before and after, and what performance indicators matter for your applications.



Verify & Prevent Hallucinations

Don't accept 'everything looks good'. Ask for proof: startup logs, endpoint tests, latency comparisons. Compare performance with your previous release. A human must approve before going live.



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How to Make Your AI a Deployment Expert

Teach It the Process

Tell your AI what success looks like: which services to deploy, acceptable downtime, and validation criteria. Clear instructions produce better results.



Give It Real Data

Connect your AI to your Huawei Cloud account and CodeArts documentation. It should know your actual pipelines, environments, and resource options -- not guess or assume.



Set Clear Expectations

Install deployment skills from our GitHub repository. Point your AI to Huawei Cloud CodeArts and KooCLI documentation. Works with OpenCode, Hermes, or Claude Code.



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Why AI-Assisted Deployment

Speed & Efficiency

Planning that used to take days now takes hours. AI discovers 100+ repositories and pipelines in minutes. 40% faster releases with automated orchestration.



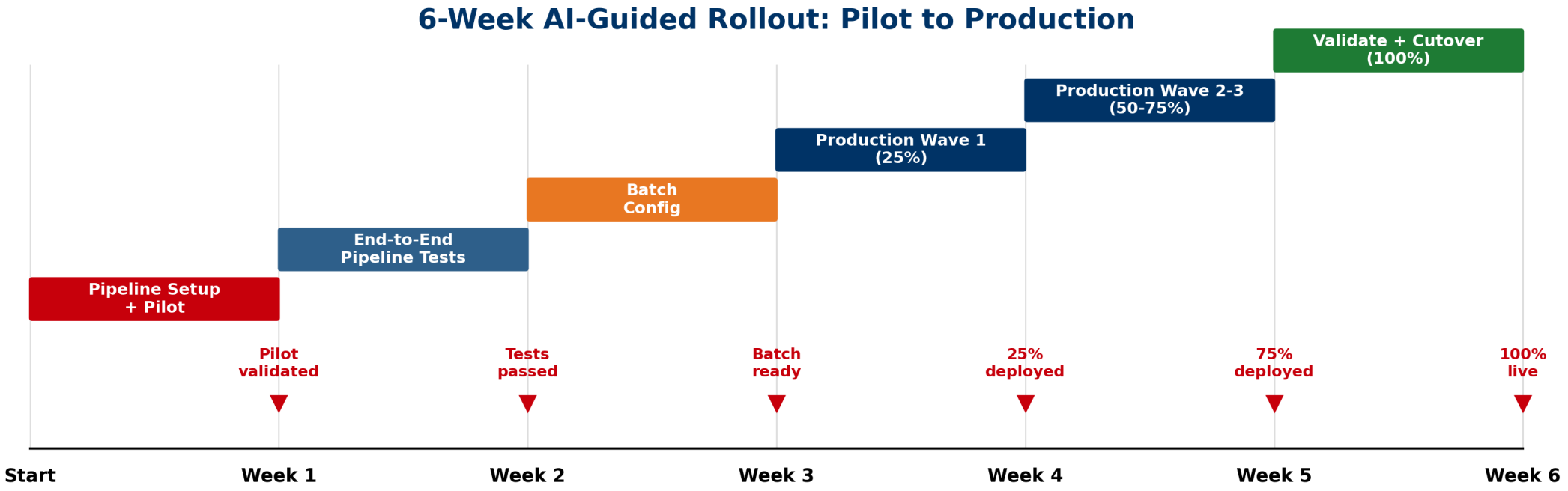
Risk Reduction

AI spots configuration issues and dependencies early. If a rollout won't succeed one way, it automatically tries another. Over 95% succeed on the first try.



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Your 6-Week AI-Guided Rollout



Week 1 -- Example prompt: "Set up the pipeline and deploy 2-3 non-critical services to the test environment. Confirm everything works."
Week 2 -- Example prompt: "Test the full CI/CD pipeline end-to-end, from build to deploy. Fix any issues."
Week 3-5 -- Example prompt: "Roll out in waves -- 25% at a time, by service. Verify each wave before proceeding."
Week 6 -- Example prompt: "Run final checks. Switch traffic to the new release. Retire the old deployment."

Source: Huawei Cloud CodeArts Best Practices (https://support.huaweicloud.com/codearts/) — Continuous Delivery Guide. Accessed: Aug 2026.

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Equip Your AI Assistant

Install Deployment Skills:
https://github.com/binrogithub/1-3-Cloud-Adoption-Skills/tree/main/AI-Assist-Migration/CodeArts-KooCLI-Deployment

AI Tools: OpenCode, Hermes, or Claude Code

Getting Started:
1. Install skills in ~/.opencode/skills/
2. Open your AI coding tool
3. Example prompt: "Help me deploy my CodeArts project to Huawei Cloud using KooCLI"

Skills include: step-by-step guides, automation scripts, and runbooks

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